

Work began in three sections or “legs” as it was called, each throwing up its own peculiar geographical challenges, with insurgency taking its own toll on the construction.

The first leg, 25 km long, connected Udhampur to Katra, a small town popular with Hindu devotees and the starting point of a trek up to the holy temple of Vaishno Devi. Scheduled to open in 2007, it involved the construction of 11 km of tunnels, 36 major and minor bridges, and traversing sharp curves. This included the country’s longest steel girder bridge over the River Jhajjar, 85-m-high. There were substantial delays because of extensive seepage and the swelling of soil, where soil increases or shrinks according to its moisture content. A subsequent tunnel collapse meant that extensive survey and redesigning was required to be done before construction commenced. The line finally opened in 2014.

There have been several difficulties in the second leg of construction, which began in 2008. The iconic Pir Panjal Tunnel, which became operational in 2013, belongs to the complicated and daunting second leg – the 129-km-long Katra–Qazigund section. At 11 km, this is the longest transportation tunnel in the country. The tunnel is perhaps one of the trickier sections with the highest “overburden” of soil strata of 1,100 m in the country. Straight and running in a north–south direction, the tunnel is 440 m lower than the road tunnel, making it less vulnerable to snow.

Author and Indian Railways expert, Vinoo Mathur in his book *Bridges, Buildings and Black Beauties*, notes that the route is extremely challenging because “the mountain is young, geologically unstable, the rock structure extremely fragile and the line has to be built across four different thrust zones”. That is why, he writes, “each stretch of the line provides a new challenge to the railway engineer.” In fact, 108 km of the track is expected to pass through tunnels with remaining lines crossing at least 70 bridges, including the record-breaking [Chenab Bridge](#). Work is still underway and the new target for completion is 2021.

The designated third leg of the stretch, the 118-km-long Qazigund–Baramulla section in the Kashmir Valley, began operations far earlier, in 2008. A Diesel-electric Multiple Unit or DEMU presently runs between Banihal and Baramulla via Srinagar.